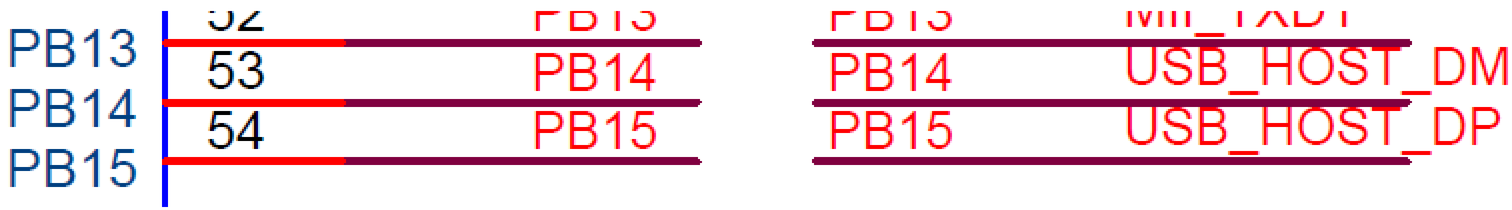
USB HOST MSC 테스트

CRZ Technology

Mango-M32F2 USB Host Mass Storage Class 테스트

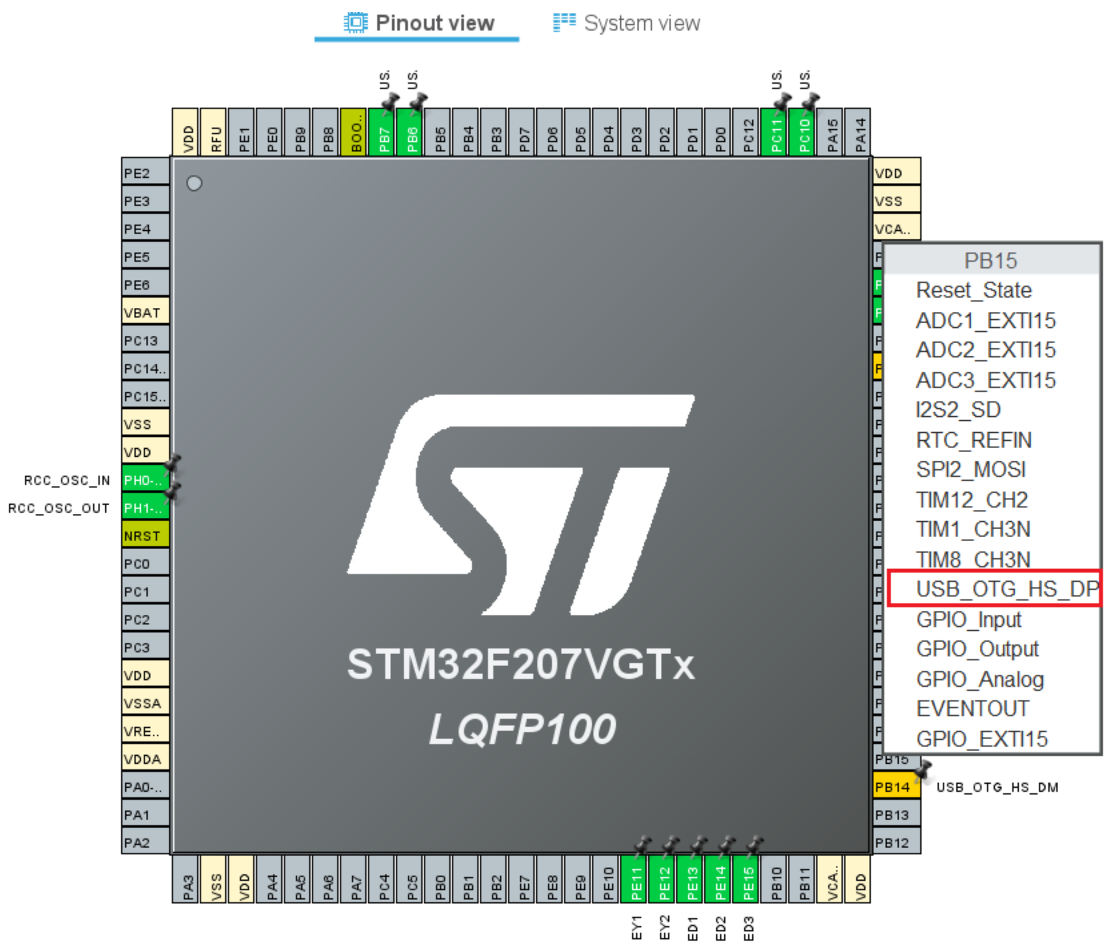
USB Host pinmap

- **USB Host – PB14 / PB15**



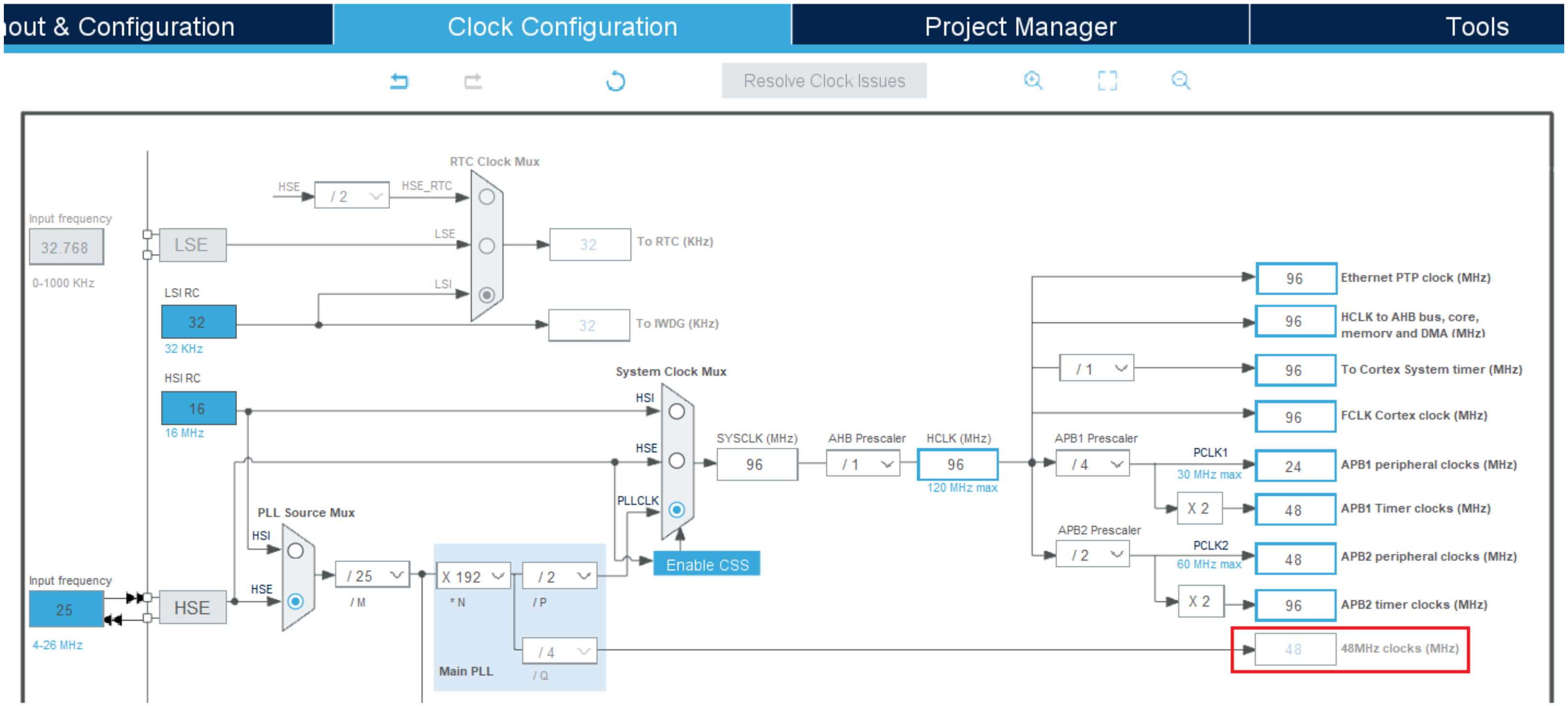
Pinout view

- **PB14 – USB_OTG_HS_DM**
- **PB15 – USB_OTG_HS_DP 선택.**



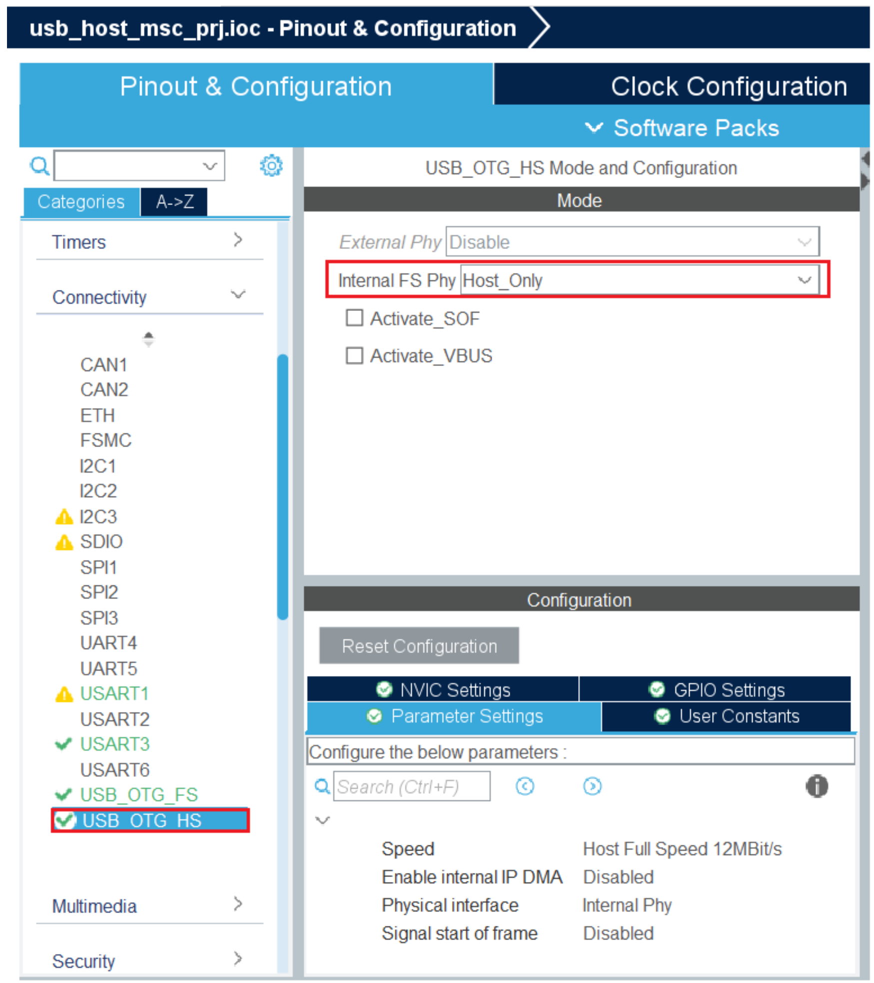
Clock 설정

■ **USB 클락 48MHz 생성되도록 설정.**



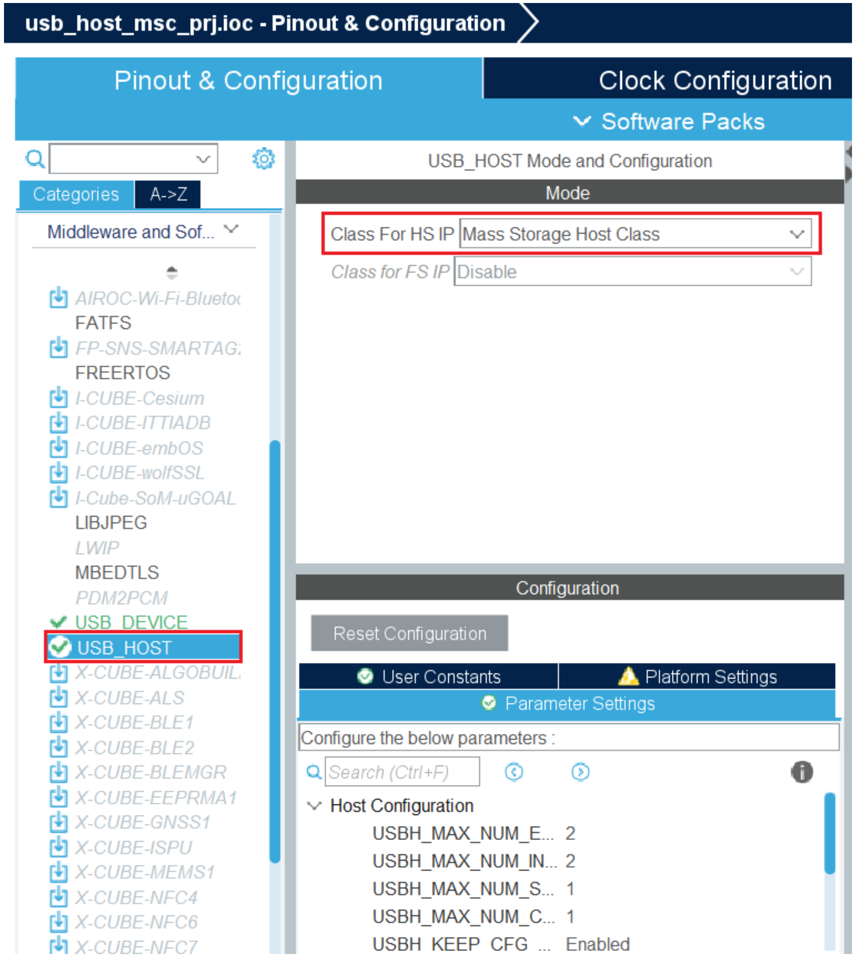
USB_OTG_HS 설정

- **“Connectivity“ 클릭, “USB_OTG_HS” 클릭.**
- **“Internal FS Phy"에서 “Host_Only” 선택.**



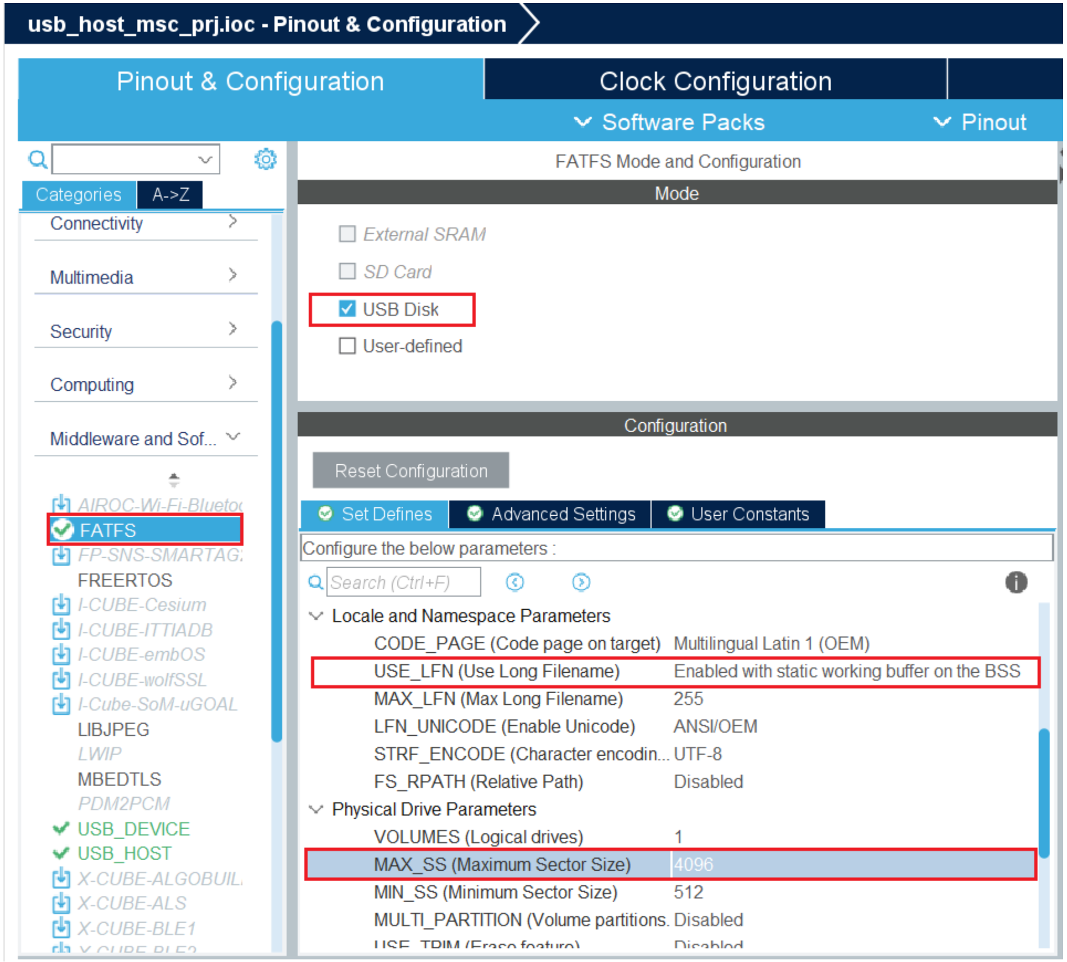
USB_HOST 설정

- "Middleware and Software Packs" 클릭, "USB_HOST" 클릭.
- "Class For HS IP"에서 "Mass Storage Host Class" 선택.



FATFS 설정

- "Middleware and Software Packs" 클릭, "FATFS" 클릭.
- "USB Disk" 클릭, "USE_LFN"에서 "Enabled with static working buffer on the BSS" 선택
- "MAX_SS"에서 "4096" 선택.



USB Host Mass Storage Class Test Project

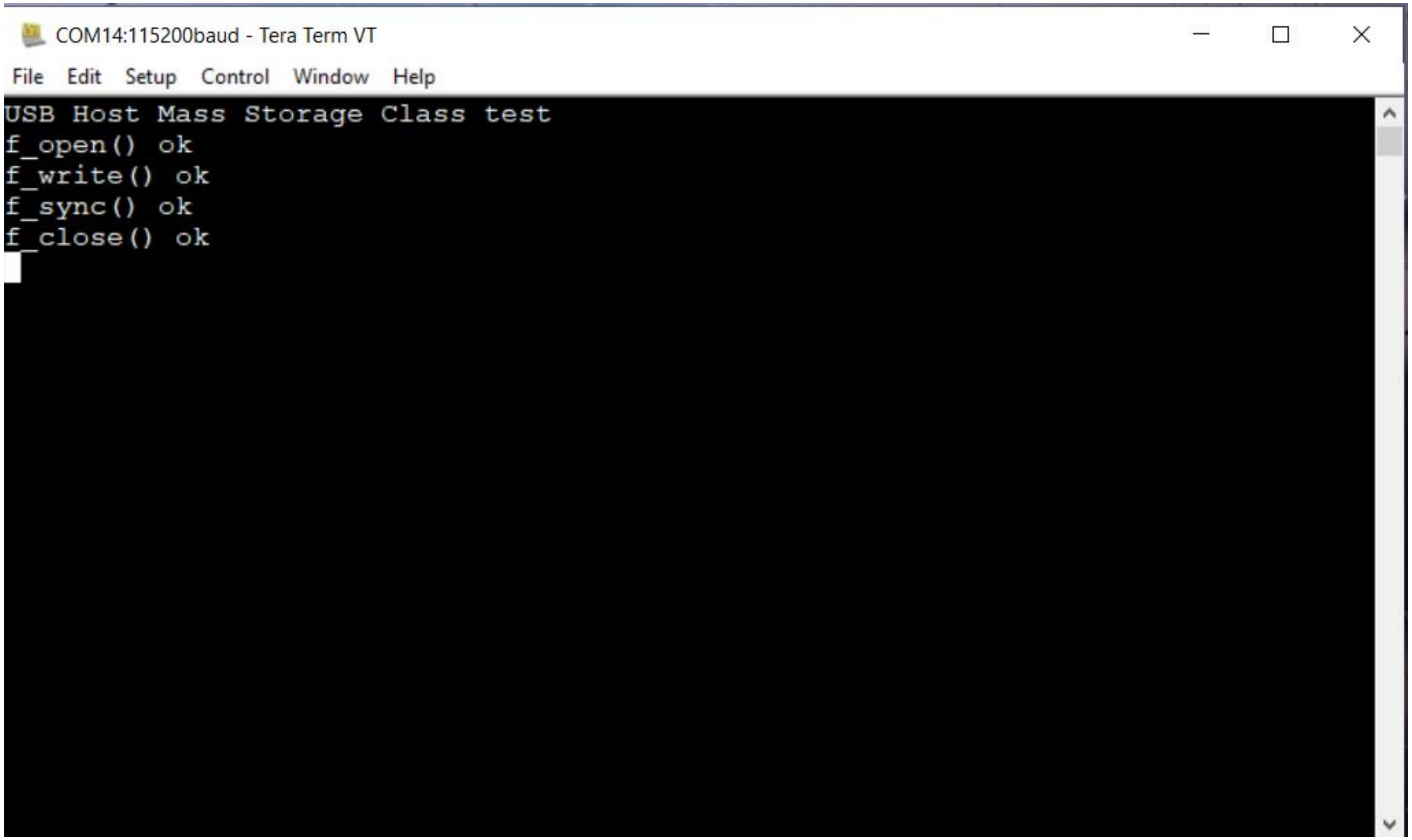
- Code 생성 후 USB Mass Storage에 파일 생성 구문 추가.
- USB_HOST/App/usb_host.c

```c
static void USBH_UserProcess  (USBH_HandleTypeDef *phost, uint8_t id)
{
  /* USER CODE BEGIN CALL_BACK_1 */
  switch(id)
  {
  case HOST_USER_SELECT_CONFIGURATION:
  break;

  case HOST_USER_DISCONNECTION:
  Appli_state = APPLICATION_DISCONNECT;
  break;

  case HOST_USER_CLASS_ACTIVE:
  Appli_state = APPLICATION_READY;

  {
    char USBHPath[4];    /* USBH logical drive path */
    FATFS USBHFatFS;     /* File system object for USBH logical drive */
    FIL USBHFile;        /* File object for USBH */
    FRESULT res;
    uint32_t byteswritten;

    char wtext[100] = "This line has been written by an STM32 device!\n";
    if(f_mount(&USBHFatFS, (TCHAR const*) USBHPath, 0) == FR_OK)
    {
        res = f_open(&USBHFile, "STM32.TXT", FA_CREATE_ALWAYS | FA_WRITE);
        if (res != FR_OK) { printf("f_open() fail, res=%d\r\n", res);  return; }
        printf("f_open() ok\r\n");

        res = f_write(&USBHFile, wtext, strlen(wtext), (void *)&byteswritten);
        if (res != FR_OK) { printf("f_write() fail, res=%d\r\n", res);  return; }
        printf("f_write() ok\r\n");

        res = f_sync(&USBHFile);
        if (res != FR_OK) { printf("f_sync() fail, res=%d\r\n", res);  return; }
        printf("f_sync() ok\r\n");

        res = f_close(&USBHFile);
        if (res != FR_OK) { printf("f_close() fail, res=%d\r\n", res);  return; }
        printf("f_close() ok\r\n");
    }
  }
  break;
```

USB Host Mass Storage Class Test Project

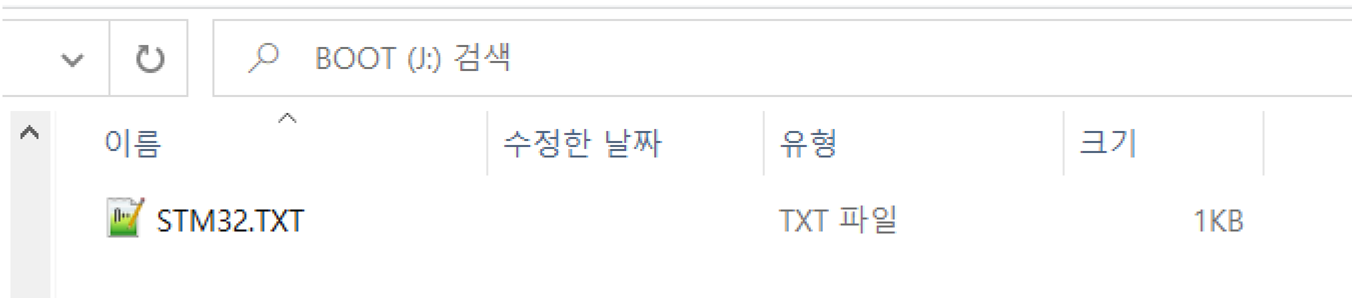
- 이미지 빌드 후 다운로드하고 부팅.
- USB Mass Storage를 CON8에 부착.

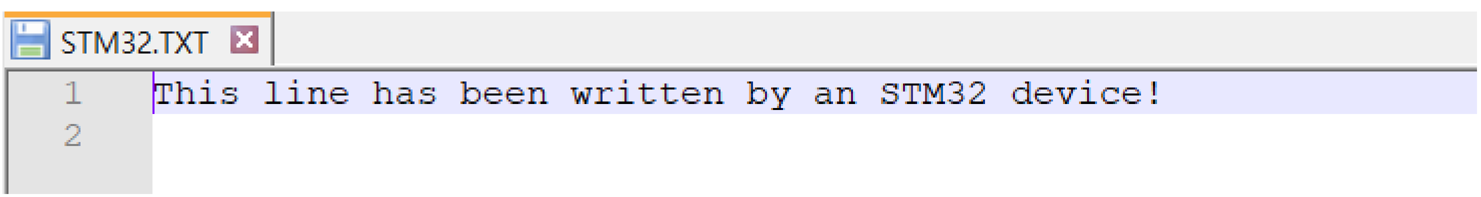
Test 결과 확인


COM14:115200baud - Tera Term VT
File Edit Setup Control Window Help
USB Host Mass Storage Class test
f_open() ok
f_write() ok
f_sync() ok
f_close() ok

Test 결과 확인

■ **USB Mass Storage를 PC에 연결 후 확인.**





Thank You